

1. Check the Change Request

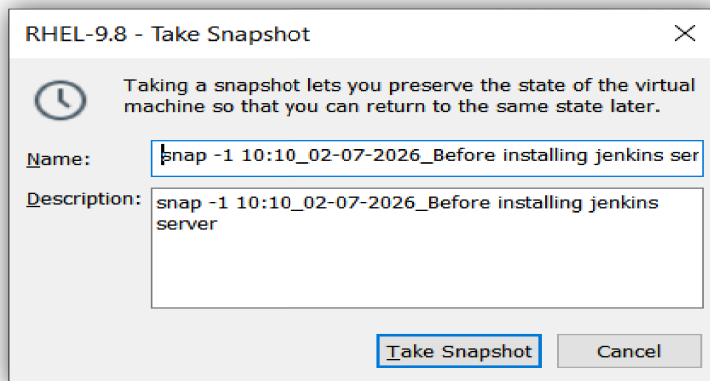
- Check whether the change is assigned to you by the TL or manager.
- Check the change status:
 - **Approved**
 - Pending
- If the change is approved, then start the work.

2. Access the Server

- Take the server IP from the TL.
- Login using:
 - **SSH**
 - RDP
 - VNC
 - Remote Console

3. Take Backup

- Always take backup before making changes.
- Backup types:
 - Physical Server → NetBackup (Full OS Backup)
 - **Virtual Machine → Snapshot Backup**



- AWS Instance → Snapshot Backup

4. Configure Network and Hostname

- **Network Adopter:** - NAT
- **Configure Static IP:-**
[root@dhcpsrv1 ~]# nmcli con mod ens33 ipv4.method auto
[root@dhcpsrv1 ~]# nmcli con up ens33

```
[root@localhost ~]# ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.5.146 netmask 255.255.255.0 broadcast 192.168.5.255
    inet6 fe80::20c:29ff:fed0:9028 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:d0:90:28 txqueuelen 1000 (Ethernet)
    RX packets 348 bytes 49253 (48.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 365 bytes 54234 (52.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- **Set the hostname:** -

```
[root@localhost ~]# hostnamectl set-hostname jenkins.radical.com
```

- **Check the hostname :-**

```
[root@localhost ~]# hostname
```

- **Configure local DNS in /etc/hosts.**

```
[root@jenkins ~]# vi /etc/hosts
```

```
[root@jenkins ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.5.146 jenkins.radical.com      jenkins
```

- **Test IP, hostname, and internet connectivity.**

```
[root@jenkins ~] # ping 192.168.5.146
```

```
[root@jenkins ~] # ping www.google.com
```

```
[root@jenkins ~] # ping www.redhat.com
```

```
[root@jenkins ~]# ping 192.168.5.146
PING 192.168.5.146 (192.168.5.146) 56(84) bytes of data.
64 bytes from 192.168.5.146: icmp_seq=1 ttl=64 time=0.069 ms
64 bytes from 192.168.5.146: icmp_seq=2 ttl=64 time=0.115 ms
64 bytes from 192.168.5.146: icmp_seq=3 ttl=64 time=0.120 ms
^C
--- 192.168.5.146 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2081ms
rtt min/avg/max/mdev = 0.069/0.101/0.120/0.022 ms
[root@jenkins ~]#
[root@jenkins ~]#
[root@jenkins ~]# ping www.google.com
PING www.google.com (142.251.150.119) 56(84) bytes of data.
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=1 ttl=128 time=10.4 ms
64 bytes from 142.251.150.119 (142.251.150.119): icmp_seq=2 ttl=128 time=21.1 ms
^C
--- www.google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1010ms
rtt min/avg/max/mdev = 10.436/15.746/21.056/5.310 ms
[root@jenkins ~]#
[root@jenkins ~]#
[root@jenkins ~]# ping www.redhat.com
PING e3396.dscx.akamaiedge.net (23.38.58.41) 56(84) bytes of data.
64 bytes from a23-38-58-41.deploy.static.akamaitechnologies.com (23.38.58.41): icmp_seq=1 ttl=128 time=30.4 ms
64 bytes from a23-38-58-41.deploy.static.akamaitechnologies.com (23.38.58.41): icmp_seq=2 ttl=128 time=43.4 ms
^C
--- e3396.dscx.akamaiedge.net ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1009ms
rtt min/avg/max/mdev = 30.387/36.871/43.355/6.484 ms
```

5. Configure Repository

- **Local Repository**

```
[root@jenkins ~]# mount /dev/sr0 /mnt
```

```
[root@jenkins ~] # vi /etc/yum.repos.d/local.repo
```

```
[AppStreamrepo]
name=this is AppStream file base repo
baseurl=file:///mnt/AppStream
enabled=1
gpgcheck=0

[BaseOS]
name=this is BaseOS file base repo
baseurl=file:///mnt/BaseOS
enabled=1
gpgcheck=0
```

- RHN Repository
- EPEL Repository
- Third-party Repository
- Satellite Server Repository

6. Verify and Install Required Packages :-

- Jenkins server already configured (installed :- java-21,maven,git, Jenkins-core)
- go to browser on jenkins server (Login)
 - * go to setting
 - * click on system
 - * go down
 - * click on add ssh

SSH Servers

SSH Server

Name ?

ssh

Hostname ?

192.168.15.147

Username ?

root

Remote Directory ?

/opt

☐ Avoid sending files that have not changed ?

Advanced ^

Edited

- click on advance

Advanced ^

Edited

☒ Use password authentication, or use a different key ?

Passphrase / Password ?

Success

Test Configuration

- **Apply and save**

- Now, create item (job)

Git ?

Repositories ?

Repository URL ?

`https://github.com/PravinA143/adnan-webapp.git`

Credentials ?

- none -

+ Add

Branch :- main

Poll SCM :- * * * * *

Goals and options :- clean install

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Send build artifacts over SSH ?

SSH Publishers

SSH Server

Name ?

ssh

Advanced

Transfers

Transfer Set

Source files ?

target/*.war

Remove prefix ?

target

Remote directory ?

//opt

Apply and save

- Click on build now ----- > output --- > Success

Now, go to another machine and configure docker_tomcat-container

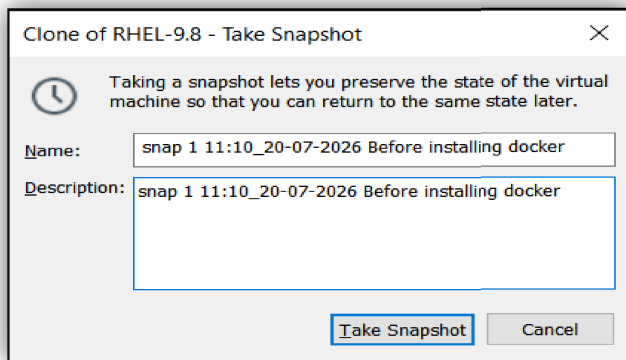
1. Access the Server

- Take the server IP from the TL.
- Login using:
 - SSH
 - RDP
 - VNC
 - Remote Console

2. Take Backup

- Always take backup before making changes.
- Backup types:
 - Physical Server → NetBackup (Full OS Backup)

- **Virtual Machine → Snapshot Backup**



- **AWS Instance → Snapshot Backup**

3. Configure Network and Hostname

- **Network Adopter:** - NAT
- **Configure Static IP:-**

```
[root@dhcp1srv1 ~]# nmcli con mod ens33 ipv4.method auto
[root@dhcp1srv1 ~]# nmcli con up ens33
```

```
[root@localhost ~]# ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.5.147 netmask 255.255.255.0 broadcast 192.168.5.255
    inet6 fe80::20c:29ff:fe19:dc43 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:19:dc:43 txqueuelen 1000 (Ethernet)
    RX packets 992 bytes 104088 (101.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 643 bytes 82902 (80.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

- **Set the hostname:** -

```
[root@localhost ~] # hostnamectl set-hostname docker.container.tomcat.radical.com
```

- **Check the hostname :-**

```
[root@docker ~] # hostname
```

- **Configure local DNS in /etc/hosts.**

```
[root@docker ~] # vi /etc/hosts
```

```
[root@docker ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.5.147  docker.radical.com    docker
```

- **Test IP, hostname, and internet connectivity.**

```
[root@docker ~] # ping 192.168.5.147
```

```
[root@docker ~] # ping www.google.com
```

```
[root@docker ~] # ping www.redhat.com
```

- **Verify/Search/Install packages :-**

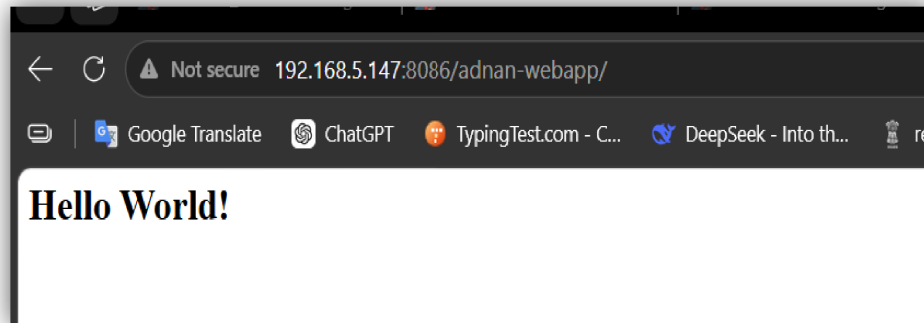
First install java & tomcat

```
# yum search java
```

```
# yum install -y java-21-openjdk.x86_64
```

```
# java --version
# wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.57/bin/apache-tomcat-10.1.57.tar.gz
# tar -xvf apache-tomcat-10.1.57.tar.gz
# mv apache-tomcat-10.1.57 tomcat
# mv apache-tomcat-10.1.57.tar.gz /tmp/
# cd /opt/tomcat/bin
# ./startup.sh
# cp -a /opt/adnan-webapp.war /opt/tomcat/webapps/
```

- Now go to browser and put ip_address+port_number+web-app_name(192.168.5.147:8080/adnan-webapp)



- Now, install docker follow the steps

```
# dnf remove -y docker \
    docker-client \
    docker-client-latest \
    docker-common \
    docker-latest \
    docker-latest-logrotate \
    docker-logrotate \
    docker-engine \
    podman \
    runc
# dnf config-manager --add-repo https://download.docker.com/linux/rhel/docker-ce.repo
# dnf install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
# systemctl enable --now docker.service
# docker pull tomcat
# docker images
# cd /opt
# vi dockerfile
```

```
[root@docker ~]# cat dockerfile
FROM tomcat:latest
MAINTAINER adnan
RUN cp -R /usr/local/tomcat/webapps.dist/* /usr/local/tomcat/webapps
COPY ./*.war /usr/local/tomcat/webapps
```

```
# docker build -t tomcat:fixed .
# docker images
# docker run -d --name web-c1 -p 8086:8080 tomcat:fixed
# docker ps
# go to browser search --> http://192.168.5.147:8086 and http://192.168.5.:8086/adnan-webapp)
```

go to jenkins server install and configure the ansible-core

- **First create ssh_trust**
ssh-keygen -t rsa
ssh-copy-id 192.168.5.147 -----> docker_Server ip
ssh 192.168.5.147 -----> Verify

Now, install ansible package

```
# yum search ansible
# yum install -y ansible-core.x86_64
# vi /etc/ansible/hosts
```

```
##### below setting done by adnan #####
[docker_srv]
192.168.5.147
```

ansible all -m ping

```
[root@jenkins .ssh]# ansible all -m ping
192.168.5.147 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
[root@jenkins .ssh]#
```

```
# cd /etc/ansible/
# pwd ---> /etc/ansible/
# vi mydockerplaybook.yml
```

```
- name: Build image and deploy image in container
  hosts: all
  become: yes
```

tasks:

stop docker container

- name: stop existing docker container

command: docker stop web-c1

clean and delete stopped docker container

- name: delete docker container

command: docker rm web-c1

clean and delete stopped docker image

- name: delete docker image

command: docker rmi tomcat:fixed

build docker image

- name: build docker image

command: docker build -t tomcat:fixed /opt/ -f /opt/dockerfile

start Docker container with new image

- name: start new docker container

command: docker run -d --name web-c1 -p 8086:8080 tomcat:fixed

ansible-playbook mydockerplaybook.yml --syntax-check

```
[root@jenkins ansible]# ansible-playbook mydockerplaybook.yml --syntax-check
playbook: mydockerplaybook.yml
```

ansible-playbook mydockerplaybook.yml -C ----- > dry run

ansible-playbook mydockerplaybook.yml

```
[root@jenkins ansible]# ansible-playbook mydockerplaybook.yml
PLAY [Build image and deploy image in container] *****
TASK [Gathering Facts] *****
ok: [192.168.5.147]
TASK [stop existing docker container] *****
changed: [192.168.5.147]
TASK [delete docker container] *****
changed: [192.168.5.147]
TASK [delete docker image] *****
changed: [192.168.5.147]
TASK [build docker image] *****
changed: [192.168.5.147]
TASK [start new docker container] *****
changed: [192.168.5.147]
PLAY RECAP *****
192.168.5.147 : ok=6 changed=5 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
```

- go to browser jenkins server

go to setting ----- > plugins ----- > install ansible plugin

🔍 ansible

Name ↓

[Ansible plugin](#) 635.v34b_48e979c86

Invoke [Ansible Ad-Hoc](#) commands and playbooks.

[Report an issue with this plugin](#)

restart the jenkins (systemctl restart jenkins)

go to setting --- > tools
* go down

Ansible installations

+ Add Ansible

Ansible

Name

ansible

Path to ansible executables directory

/usr/bin

☐ Install automatically ?

+ Add Ansible

Save Apply

go to setting ---- > credentials
* add credentials
* Select a type of credential :- username with password

< Add Username with password ×

Scope ?

Global (Jenkins, nodes, items, all child items, etc) ▼

Username

root

☐ Treat username as secret ?

Password

redhat 🔍

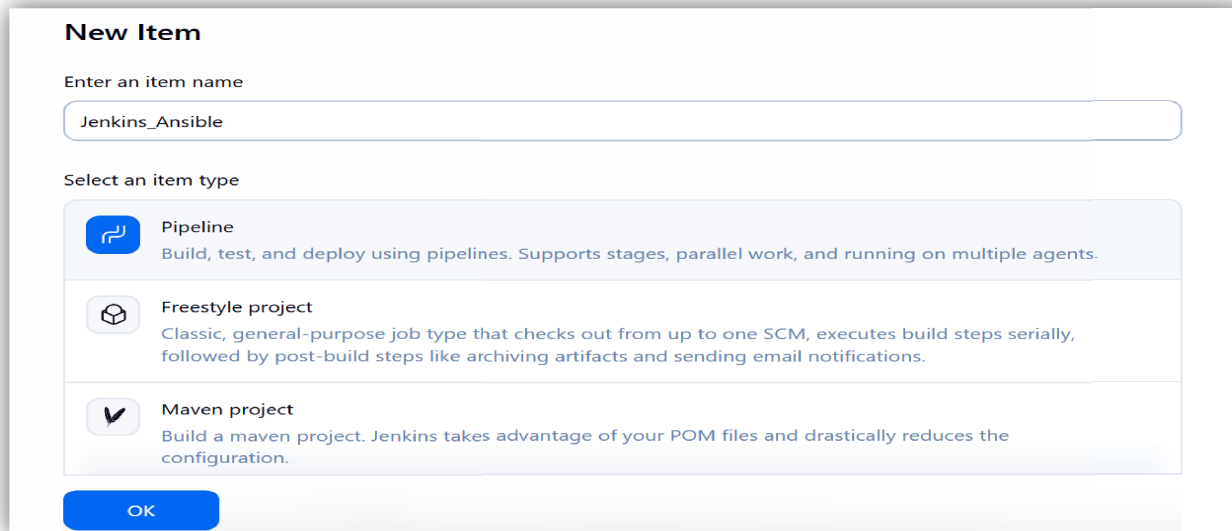
ID ?

Description ?

Create

** setting ---- > system ----- > ssh_server ----- > already created

** create new item (job)



The 'New Item' dialog box in Jenkins. It has a title 'New Item'. Below it is a text input field labeled 'Enter an item name' with the value 'Jenkins_Ansible'. Below that is a section 'Select an item type' with three options: 'Pipeline' (selected), 'Freestyle project', and 'Maven project'. Each option has a description. At the bottom is an 'OK' button.

New Item

Enter an item name

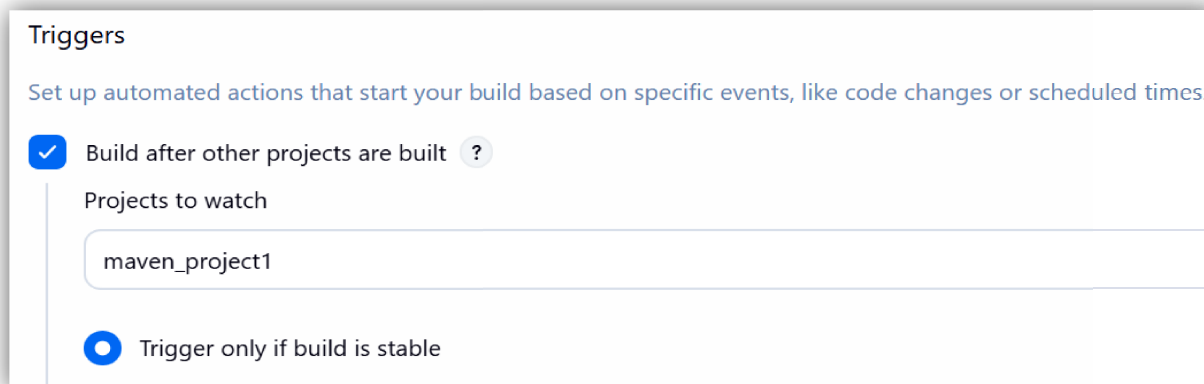
Jenkins_Ansible

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

OK

Build after other projects are built :- maven_project1----- > this name is a maven_ssh project name



The 'Triggers' configuration section in Jenkins. It has a title 'Triggers' and a description 'Set up automated actions that start your build based on specific events, like code changes or scheduled times.' Below this is a checkbox 'Build after other projects are built' which is checked. To its right is a help icon. Below this is a text input field labeled 'Projects to watch' with the value 'maven_project1'. At the bottom is a radio button 'Trigger only if build is stable' which is selected.

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☒ Build after other projects are built ?

Projects to watch

maven_project1

☒ Trigger only if build is stable

Pipeline definition : - select Pipeline script

```
pipeline{
  agent any
  stages{
    stage("Execute playbook"){
      steps{

      }
    }
  }
}
```

Definition

Pipeline script

Script ?

```

1 pipeline{
2   agent any
3   stages{
4     stage("Execute playbook"){
5       steps{
6
7       }
8     }
9   }
10 }
11 }

```

try sample Pipeline...

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

click on :- Pipeline Syntax

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Sample Step select --- > ansibleAdhoc: Invoke an ansible adhoc command

Steps

Sample Step

ansiblePlaybook: Invoke an ansible playbook

ansiblePlaybook ?

Ansible tool

ansible

Playbook file path in workspace

/etc/ansible/mydockerplaybook.yml

Inventory file path in workspace

/etc/ansible

SSH connection credentials

root /*****

+ Add

click on generate pipeline script

Generate Pipeline Script

ansiblePlaybook credentialsId: '144de1c4-85af-465f-9cc8-b58591efd2a4', installation: 'ansible', inventory: '/etc/ansible', playbook: '/etc/ansible/mydockerplaybook.yml', vaultTmpPath: "



Then copy the script and go back
paste the line number 6

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 pipeline{
2     agent any
3     stages{
4         stage("Execute playbook"){
5             steps{
6                 ansiblePlaybook credentialsId: '144de1c4-85af-465f-9cc8-b58591efd2a4', installation: 'ansible', inventory: '/etc/ansible', playb
7             }
8         }
9     }
10 }
11 }
```

try sample Pipeline...

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

click on Apply and Save
click on build now

if you will get this error :--

```
PLAY [Build image and deploy image in container] *****

TASK [Gathering Facts] *****
fatal: [192.168.5.147]: FAILED! => {"msg": "Using a SSH password instead of a key is not possible because Host Key checking is enabled and sshpass does not support this. Please add this host's fingerprint to your known_hosts file to manage this host."}

PLAY RECAP *****
192.168.5.147      : ok=0   changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0

FATAL: command execution failed
hudson.AbortException: Ansible playbook execution failed
```


** then configure the ansible configuration file

vi /etc/ansible/ansible.cfg

[defaults]

host_key_checking = False

```
# Since Ansible 2.12 (core):
# To generate an example config file (a "disabled" one with all default settings, commented out):
# $ ansible-config init --disabled > ansible.cfg
#
# Also you can now have a more complete file by including existing plugins:
# ansible-config init --disabled -t all > ansible.cfg
#
# For previous versions of Ansible you can check for examples in the 'stable' branches of each version
# Note that this file was always incomplete and lagging changes to configuration settings
# for example, for 2.9: https://github.com/ansible/ansible/blob/stable-2.9/examples/ansible.cfg

[defaults]
host_key_checking = False
```

go to browser jenkins server ----- > build Now --- > Successful

go to browser access webapp :- 192.168.5.147:8086/adnan-webapp

Now, testing the CICD

go to Jenkins server and edit the index.jsp

cd /root/adnan-webapp

vi src/main/webapp/index.jsp

```
<html>
<body>
<h2><%= "Hello World! i have completed CICD pipeline and testing successfull! " %></h2>
</body>
</html>
```

git status

git add .

git commit -m "code1 modified for testing"

git push -u origin main

Username :- adnan

Password :- <token_number>

go to web page only refresh it



Not secure 192.168.5.147:8086/adnan-webapp/#gallery



Google Translate



ChatGPT



TypingTest.com - C...



DeepSeek - Into th...



register.eshram.go...



Hello World! i have completed CICD pipeline and testing successfull!